

Literature Review: Auditing Geographic and Cultural Bias in Clinical Large Language Models

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Abstract

This literature review surveys fourteen papers at the intersection of clinical artificial intelligence, algorithmic fairness, and large language model (LLM) evaluation, collectively motivating a systematic audit of geographic and cultural bias in clinical LLMs. The reviewed work is organised into four thematic categories: (1) racial and sociodemographic bias in clinical AI systems; (2) geographic and cultural bias in language models; (3) clinical LLM evaluation frameworks and benchmarks; and (4) bias-detection methodologies based on perturbation and counterfactual analysis. The literature shows that LLMs consistently produce disparate clinical recommendations driven by non-clinical identity cues—patient name, race, dialect, and geographic origin—even when the underlying medical presentation is held constant. Crucially, however, every reviewed clinical bias study operates within Global-North demographic categories; the Global North/South axis has never been evaluated as a systematic perturbation dimension. This review identifies that gap as the central motivation for our project, which introduces the Geographic Disparity Index (GDI) as a novel fairness metric and applies semantic-preserving geographic perturbation across seven LLMs and four clinical datasets to quantify geographic bias in AI-assisted clinical decision-making.

1 Introduction

Large Language Models (LLMs) are increasingly being deployed or considered for deployment in clinical decision-support roles, particularly in regions facing acute physician shortages [19]. In the Global South—encompassing South Asia, Sub-Saharan Africa, Southeast Asia, the Middle East, and Latin America—over six billion people increasingly interact with or are targeted by AI-assisted clinical tools, presenting both enormous promise and substantial risk. The promise lies in democratising access to medical guidance; the risk lies in whether models trained predominantly on English-language Western biomedical text encode systematic biases that disadvantage patients from underrepresented geographic and cultural backgrounds.

This research area sits at the confluence of *big data analytics*, *natural language processing*, and *health equity*. From a data perspective the scale of the problem is immense: modern LLMs are trained on corpora containing billions of tokens, and even small systematic asymmetries in training data—for example, an over-representation of patients named “David from New York” relative to “Ali from Karachi”—can be amplified at inference time across millions of clinical interactions globally. Detecting and quantifying such biases requires large-scale experimental designs involving tens of thousands of perturbation trials across multiple models, datasets, and demographic dimensions—precisely the kind of task for which big-data pipelines are indispensable.

The specific problem we address is: *Do clinical LLMs recommend different levels of care for patients with identical medical presentations when the patient’s geographic or cultural identity is varied through name and location signals?* This is a health-equity question with direct societal consequences: a model that systematically under-refers Global South patients for specialist evaluation or diagnostic workup reproduces structural inequities at algorithmic scale with an appearance of neutrality.

The objectives of this literature review are: (1) to survey existing evidence for demographic and geographic bias in clinical AI systems; (2) to review methodologies for bias detection in LLMs, particularly perturbation-based approaches; (3) to assess clinical LLM evaluation frameworks relevant to our experimental design; and (4) to identify the research gaps our project is positioned to fill. The review is organised as follows: Section 2 provides background and formal definitions; Section 3 surveys the literature in four thematic categories; Section 4 provides comparative analysis; Section 5 identifies research gaps; Section 6 articulates the connection to our project; and Section 7 concludes.

2 Background and Preliminaries

2.1 Key Definitions

Large Language Model (LLM). A transformer-based neural network trained on large text corpora via next-token prediction [13]. In clinical contexts LLMs are prompted with patient descriptions, symptoms, and queries to generate free-text management recommendations.

Algorithmic Bias. A systematic and unjustified deviation in model outputs for members of different demographic groups [9]. Bias may arise from training data imbalance, proxy variables, or the implicit encoding of societal disparities.

Demographic Perturbation. An evaluation technique in which non-clinical identity signals (e.g., patient name, city of origin) are varied while clinical content is held constant, to test whether model outputs change [1, 10].

Semantic-Preserving Perturbation. A specific variant in which all medically relevant tokens—symptoms, history, medications, examination findings—are held strictly constant; only the identity-signalling tokens are modified.

Global North / Global South. A socioeconomic and geopolitical distinction between historically industrialised nations (US, UK, Europe, Australia) and developing nations (South Asia, Sub-Saharan Africa, Southeast Asia, MENA, Latin America).

Clinical Decision Support (CDS). AI- or algorithm-assisted guidance for medical

practitioners or patients about diagnosis, treatment, or care escalation.

2.2 Foundational Techniques

Transformer Architecture and Attention. LLMs encode contextual information through multi-head self-attention. The attention between position i and position j is:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V \quad (1)$$

where Q, K, V are query, key, and value matrices and d_k is the key dimension. Name tokens and geographic tokens may attract disproportionate attention, causing identity signals to modulate clinical reasoning [3].

LLM-as-Annotator. Given the scale of output annotation required ($\sim 75,000$ model responses), human annotation is infeasible. Following Zheng et al. [7] we use a secondary LLM to extract structured binary judgements from model outputs. For a model response r , the annotator extracts:

$$y_{\text{MANAGE}} = \mathbf{1}[\text{response recommends home self-management}] \quad (2)$$

$$y_{\text{VISIT}} = \mathbf{1}[\text{response recommends clinic or ED visit}] \quad (3)$$

$$y_{\text{RESOURCE}} = \mathbf{1}[\text{response recommends diagnostic resource}] \quad (4)$$

2.3 Formal Problem Statement and Novel Metrics

Let $\mathcal{C}$ be a set of clinical cases with ground-truth clinician annotations. For each case $c \in \mathcal{C}$, let $x_c^{(0)}$ denote the baseline (Global North) prompt and $x_c^{(k)}$ the k -th geographically perturbed variant. Let f_θ denote an LLM with parameters θ .

Treatment Shift Rate (TSR):

$$\text{TSR} = \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} \mathbf{1}[f_\theta(x_c^{(k)}) \neq f_\theta(x_c^{(0)})] \quad (5)$$

Reduced Care Rate (RCR):

$$\text{RCR} = \frac{|\{c : \text{shift is care-reducing}\}|}{|\mathcal{C}|} \quad (6)$$

Reduced Care Error Rate (RCER):

$$\text{RCER} = \frac{|\{c : \text{shift is care-reducing} \wedge \text{contradicts clinician label}\}|}{|\{c : \text{shift is care-reducing}\}|} \quad (7)$$

Geographic Disparity Index (GDI) – our novel composite metric:

$$\text{GDI} = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} w_r \left(\text{RCER}_r^{(\text{South})} - \text{RCER}_r^{(\text{North})} \right) \quad (8)$$

where $\mathcal{R}$ is the set of geographic regions, w_r is a population-proportional weight, and $\text{RCER}_r^{(\cdot)}$ is the RCER under Global South and Global North perturbations respectively. $\text{GDI} > 0$ with $p < 0.005$ (Bonferroni-corrected) indicates systematic geographic bias.

3 Literature Survey

3.1 Category 1: Racial and Sociodemographic Bias in Clinical AI Systems

Clinical AI systems have repeatedly been shown to encode and amplify sociodemographic disparities. This section surveys four foundational papers establishing the scope and severity of this problem.

Obermeyer et al. [8]: Dissecting Racial Bias in a Health Algorithm

Problem addressed. A widely deployed commercial health-risk algorithm was suspected of systematically under-assigning risk scores to Black patients relative to equally ill White patients, leading to fewer referrals for supplemental care programmes.

Approach. The authors reverse-engineered the algorithm by analysing 43,539 primary-care patients, discovering that it used *healthcare costs* as a proxy for *health needs*—a proxy corrupted by structural racism, since Black patients historically spend less on healthcare even when equally sick due to systemic access barriers.

Key contributions. The paper provides canonical evidence that algorithmic health proxies can reproduce racial disparities without containing explicit racial features. It establishes the auditing methodology of comparing algorithm outputs across matched cohorts that differ only in race.

Dataset. 43,539 patients from a large academic medical centre; outcome labels from EHR records.

Results. At a given risk score, Black patients were found to be significantly sicker than White patients. The algorithm effectively assigned lower risk to Black patients despite higher disease burden, reducing Black enrolment in care management programmes by 50%.

Limitations. Focuses on a single proprietary algorithm at a single institution. The bias mechanism (cost as proxy) may not generalise to LLM-based systems that operate on natural language.

Zack et al. [2]: GPT-4 Perpetuates Racial and Gender Biases in Healthcare

Problem addressed. Whether GPT-4 reproduces documented racial and gender biases encoded in medical education when performing clinical vignette tasks.

Approach. The authors tested GPT-4 on four clinical vignettes known to contain erroneous race- and gender-based clinical recommendations (e.g., race-adjusted kidney function estimates), evaluating whether GPT-4 perpetuated these biases in free-text responses.

Key contributions. Direct evidence that GPT-4 reproduces race-based medical myths (e.g., spirometry race adjustment, differential creatinine correction for Black patients) encoded in historical medical texts. Establishes LLMs as potential vectors for disseminating debunked race-based pseudoscience.

Dataset. 4 clinical vignettes from established medical bias literature; 5 clinical tasks.

Results. GPT-4 incorporated race as a clinical variable in 65% of vignettes where it should not, reproducing documented racial biases in pain management, kidney function, and obstetric risk assessment.

Limitations. Limited to 4 vignettes and a single model. Does not apply counterfactual perturbation systematically. Exclusively U.S.-centric racial categories.

Omiye et al. [4]: Large Language Models Propagate Race-Based Medicine

Problem addressed. Whether four widely used LLMs (GPT-4, GPT-3.5, Bard, Llama-2) perpetuate race-based medical practices that lack biological justification.

Approach. The authors identified ten documented examples of debunked race-based medicine from the medical literature and queried each model on these topics, then evaluated response accuracy and racial bias through qualitative content analysis.

Key contributions. First systematic multi-model comparison of race-based medical bias in LLMs. Demonstrates that all tested models perpetuate at least some race-based medical falsehoods and introduces a content-analysis framework for evaluating race-based LLM outputs.

Dataset. 10 clinical topics $\times$ 4 models $\times$ 5 repetitions.

Results. 8 of 10 topics elicited at least one model perpetuating false race-based claims. GPT-3.5 performed worst; Llama-2 and GPT-4 showed better but not perfect performance.

Limitations. Binary evaluation (propagates bias yes/no) does not measure degree or clinical impact. Does not evaluate geographic or cultural origin as a perturbation axis.

Omar et al. [15]: Sociodemographic Biases in Medical Decision-Making by LLMs

Problem addressed. Whether LLMs produce different clinical management recommendations for emergency department (ED) patients based on sociodemographic characteristics when clinical presentations are held constant.

Approach. 1,000 ED cases were presented to 9 LLMs (GPT-4, Claude, Llama, Gemini), each in 32 sociodemographic variants (race, gender, insurance, housing, LGBTQIA+ status, BMI), yielding 1.7 million total outputs. Differences in diagnostic workup, specialist referral, and treatment intensity were measured.

Key contributions. Largest-scale study to date on sociodemographic bias in clinical LLM decision-making. Finds that Black, unhoused, and LGBTQIA+ patients are directed toward invasive interventions or urgent care with no clinical justification.

Dataset. 1,000 real ED cases; 32 demographic variants; 9 LLMs; 1.7 million outputs.

Results. Significant ($p < 0.001$) disparities across all tested sociodemographic dimensions. GPT-4 showed the most consistent disparities; Llama-3 and Gemini-Pro showed smaller but statistically significant effects.

Limitations. Exclusively U.S.-based sociodemographic categories. Does not test geographic origin (Global North vs. Global South), multilingual models, or biomedical-specialised models. Perturbation limited to explicitly declared demographic attributes

rather than name-based inference.

3.2 Category 2: Geographic and Cultural Bias in Language Models

While clinical AI literature has focused on within-U.S. racial disparities, a parallel body of NLP work has established that language models encode geographic and cultural biases intrinsically. This section reviews four papers establishing that LLMs are systematically underrepresentative of the Global South.

Gourabathina et al. [1]: The Medium is the Message

Problem addressed. Whether non-clinical information in patient messages—patient name, writing style, and perceived socioeconomic status—influences LLM-generated clinical responses in ways that parallel documented human provider bias.

Approach. The authors applied a semantic-preserving perturbation framework to a clinical QA dataset, varying patient names (Black-associated vs. White-associated names derived from U.S. Census data) and writing style (formal vs. informal) while holding all clinical content constant. GPT-4 was the evaluated model; a second GPT-4 instance served as the LLM-as-annotator.

Key contributions. Establishes the direct methodological template for our project. Introduces the three-question annotation schema (MANAGE / VISIT / RESOURCE) and the semantic-preserving perturbation approach we adopt and extend to global geographies.

Dataset. OncQA (cancer patient-clinician messages) and AskDocs (Reddit medical queries); ~208 total cases with name perturbation.

Results. Black-name patients received significantly more self-management recommendations (Reduced Care Rate: +12.3% vs. White-name baseline) and fewer specialist referrals. Informal writing style amplified the effect.

Limitations. Restricted to U.S.-centric Black/White name pairs. Tests only GPT-4. Does not extend to Global South geographies or multilingual models.

Hofmann et al. [3]: AI Generates Covertly Racist Decisions Based on Dialect

Problem addressed. Whether LLMs make covertly racist decisions when input text is written in African American English (AAE) versus Standard American English (SAE), even when the model does not explicitly recognise the dialect.

Approach. A dialect classifier was used to systematically vary input text between AAE and SAE across six decision-making tasks, including criminal sentencing and employment screening. Multiple LLMs were evaluated on the resulting disparate output distributions.

Key contributions. Demonstrates that linguistic form—not explicit race labels—can trigger discriminatory model behaviour, bypassing explicit racial fairness guardrails. Introduces the *covert racism* framing: models that deny racial bias while exhibiting it through linguistic proxies.

Dataset. Dialect corpus with 200+ texts per condition; 6 decision tasks; multiple LLMs.

Results. LLMs exhibited significantly more negative stereotyping for AAE texts. Effects persisted when models were explicitly instructed to ignore race.

Limitations. Focuses on U.S. dialect, not cross-national linguistic variation. Does not examine clinical domains or geographic origin.

Manvi et al. [11]: Large Language Models are Geographically Biased

Problem addressed. Whether LLMs have systematic geographic biases in knowledge and prediction, over-representing wealthy Western regions.

Approach. The geographic knowledge of GPT-4, Gemini, and Llama-2 was probed using geographic prediction tasks (e.g., estimating crop yields, poverty rates, and satellite-imagery features from text descriptions). Prediction errors were correlated with GDP, population density, and Global North/South classification.

Key contributions. First systematic study of geographic bias in LLMs across multiple dimensions. Shows that prediction errors correlate strongly with GDP ($r = -0.62$), meaning the Global South is systematically underserved by LLM geographic knowledge.

Dataset. Worldwide geographic prediction tasks across 170+ countries; GDP, HDI, and population data from the World Bank and UN.

Results. LLMs significantly over-perform on Global North regions and under-perform on Global South regions. The effect is not correctable by simple recalibration.

Limitations. Tests geographic prediction tasks (agriculture, economics) rather than clinical reasoning. Does not directly evaluate whether geographic bias transfers to healthcare decision-making.

Faisal and Anastasopoulos [12]: Geographic and Geopolitical Biases of Language Models

Problem addressed. Whether pretrained language models encode geographic and geopolitical biases in their representations, beyond surface-level language differences.

Approach. A Geographic Representation Probing Framework was developed using probing classifiers to test whether LM representations encode geographic location, geopolitical alignment, and wealth/development status, evaluated on mBERT, XLM-R, and GPT-2 across 193 countries.

Key contributions. Establishes that geographic and geopolitical biases are intrinsic to LM representations and amplified at inference time. Models cluster countries along geopolitical rather than geographic lines, reflecting training-data political framing.

Dataset. Wikipedia articles and news corpora for 193 countries; geopolitical alignment labels from UN voting records.

Results. Probing classifiers achieve 85% accuracy at predicting geopolitical bloc from LM representations. English-trained models show stronger Western bias.

Limitations. Focuses on representational bias rather than downstream task performance. Does not evaluate clinical or health-related tasks.

3.3 Category 3: Clinical LLM Evaluation Frameworks and Benchmarks

Effective clinical LLM auditing requires robust evaluation frameworks. This section surveys four papers that have developed methodologies for assessing LLM performance and safety in clinical settings.

Thirunavukarasu et al. [13]: Large Language Models in Medicine

Problem addressed. How do LLMs perform across the spectrum of clinical tasks, and what are the key risks and evaluation requirements for equitable clinical deployment?

Approach. A comprehensive narrative review of LLM performance across clinical tasks (diagnosis, treatment recommendation, documentation, patient communication) with meta-analysis of published benchmarks (MedQA, USMLE, ClinicalBench).

Key contributions. Establishes the canonical taxonomy of clinical LLM capabilities and failure modes. Identifies training-data bias as the primary risk to equitable clinical deployment. Calls for standardised evaluation frameworks using diverse patient populations.

Dataset. Review of 110+ papers; benchmarks include USMLE-MedQA, MedMCQA, PubMedQA.

Results. GPT-4 achieves passing scores on USMLE (> 70%) but performance degrades significantly for non-English clinical scenarios and diseases prevalent in the Global South (e.g., malaria, dengue, sickle-cell anaemia).

Limitations. Does not systematically evaluate geographic or cultural bias. Reviewed benchmarks are predominantly derived from U.S. and European clinical contexts.

Hager et al. [6]: Evaluation and Mitigation of LLM Limitations in Clinical Decision-Making

Problem addressed. How to systematically identify LLM failure modes in clinical decision-making, and whether mitigation strategies (chain-of-thought prompting, retrieval augmentation) address identified limitations.

Approach. A structured evaluation battery of 300 expert-curated clinical decision tasks was applied to GPT-4, Claude-2, and Llama-2, evaluating both accuracy and calibration. Mitigation strategies were then tested on identified failure modes.

Key contributions. Introduces the first framework specifically designed to evaluate LLM *failure modes* in clinical settings, distinguishing hallucination, knowledge gaps, and bias as distinct categories. Shows that chain-of-thought prompting improves accuracy but does not reduce demographic bias.

Dataset. 300 expert-curated clinical decision tasks; validated by 12 clinician annotators.

Results. All tested LLMs showed systematic overconfidence. Retrieval-augmented generation reduced hallucination but had no significant effect on demographic bias.

Limitations. Clinical tasks designed by U.S. and European clinicians. No systematic geographic perturbation.

Pfohl et al. [5]: A Toolbox for Surfacing Health Equity Harms in LLMs

Problem addressed. How to systematically identify and measure health equity harms produced by LLMs in a way that is reproducible across research groups.

Approach. The authors developed an evaluation toolbox comprising: (a) 457 clinical scenarios with paired patient descriptions varying by race, gender, age, and insurance; (b) a structured annotation rubric; and (c) reproducible disparity metrics.

Key contributions. First standardised toolbox for health-equity evaluation of clinical LLMs. Demonstrates that GPT-4, Bard, and Llama-2-70B all exhibit measurable health equity harms across multiple demographic dimensions.

Dataset. 457 clinical scenarios $\times$ 8 demographic variants; validated by 6 clinician annotators.

Results. All three tested LLMs exhibited significant demographic disparities in at least 3 of 6 clinical task categories. Insurance status and race produced the largest disparities.

Limitations. Geographic origin is not a tested dimension. All scenarios are U.S.-centric.

Poulain et al. [14]: Bias Patterns in LLMs for Clinical Decision Support

Problem addressed. Whether bias in clinical LLMs is model-specific or systematic across model families, and which protected attributes drive the largest disparities.

Approach. 8 LLMs (GPT-4, Claude-2, Llama-2-70B, Falcon-40B, and others) were evaluated across 3 clinical QA datasets using standardised patient vignettes. Demographic attributes were varied via template substitution; output distributions were compared using Kullback–Leibler divergence.

Key contributions. Shows that clinical LLM bias is not model-specific—all tested models exhibit significant disparities. Demonstrates that biomedical fine-tuned models show *larger* disparities than general-purpose models (+23% average), suggesting fine-tuning on clinical corpora amplifies rather than reduces bias.

Dataset. MedQA-USMLE, ClinicalNLP, MedPaLM evaluation sets; 3 demographic attributes $\times$ 5 values.

Results. Race and socioeconomic status produced the largest disparities (KL = 0.31 and 0.28 respectively). Fine-tuned models showed 23% larger average disparities than their base counterparts.

Limitations. Does not test geographic origin or non-English names. All demographic signals inserted via explicit attribute statements rather than implicit name-based inference.

3.4 Category 4: Bias Detection Methodologies: Perturbation and Counterfactual Analysis

The methodological foundation of demographic bias auditing in NLP relies on perturbation and counterfactual techniques. This section surveys three papers that have shaped the methodological landscape most directly relevant to our work.

Blodgett et al. [9]: Language Technology is Power

Problem addressed. Why is NLP bias research methodologically inconsistent, and what normative and empirical framework should guide it?

Approach. A critical survey of 146 NLP papers on “bias”, analysing their motivation, operationalisation, and claims. The authors argue that most papers conflate technical and normative questions without specifying whose interests are protected or what harms are prevented.

Key contributions. Seminal work establishing normative foundations for NLP bias research. Introduces the distinction between *representational* bias (stereotyping in model representations) and *allocational* bias (differential performance on downstream tasks). Argues for grounding bias research in documented, specific harms to named communities.

Dataset. 146 NLP papers reviewed; no new experimental data.

Results. Most surveyed papers do not specify which communities are harmed or how. Few ground their bias definitions in documented real-world harm.

Limitations. Methodological critique rather than experimental paper. Does not propose specific new evaluation datasets or metrics.

Wan et al. [10]: Gender Biases in LLM-Generated Reference Letters

Problem addressed. Whether LLMs encode gender bias when generating professional reference letters, and whether name-based perturbation reliably detects such bias.

Approach. Female-stereotyped names (Kelly, Jennifer) were replaced with male-stereotyped names (Joseph, David) in prompts requesting LLM-generated reference letters. Generated text was analysed for agentic vs. communal language using LIWC and word-embedding analyses across five LLMs.

Key contributions. Validates name-based perturbation as a reliable probe for demographic bias in LLM outputs. Shows that LLMs systematically describe female-named subjects with warmth/likeability language and male-named subjects with leadership/competence language even when the underlying request is identical. Establishes the methodological precedent we extend from gender to geographic identity.

Dataset. 100 prompts $\times$ 2 name conditions $\times$ 5 LLMs (ChatGPT, GPT-4, Alpaca, Dolly, Vicuna).

Results. Female-named subjects received 68% more communal language and 41% less agentic language, consistently across all tested LLMs.

Limitations. Binary gender proxy; does not address non-binary or intersectional identity. Limited to U.S.-centric names. Does not extend to geographic or cultural identity.

Parrish et al. [16]: BBQ: A Bias Benchmark for Question Answering

Problem addressed. How to systematically measure whether QA models rely on social stereotypes when answering questions about people from different demographic groups.

Approach. A 58,492-example benchmark of question–context–answer triplets was constructed in which the correct answer requires *not* relying on demographic stereotypes.

Each question has an ambiguous version (only stereotypes provide a cue) and a disambiguated version (context provides an unambiguous answer).

Key contributions. BBQ is the most widely used benchmark for measuring stereotype reliance in LLMs. Covers 9 social categories including race/ethnicity, nationality, religion, gender, and age. Provides a replicable methodology for constructing counterfactual evaluation sets directly applicable to clinical scenario design.

Dataset. 58,492 examples; 9 social categories; 3 question formats.

Results. All tested models showed above-baseline reliance on stereotypes in ambiguous contexts. Race/ethnicity and nationality categories produced the largest biases.

Limitations. Focused on QA classification rather than generative outputs. Nationality categories do not specifically distinguish Global North/South. Does not test LLMs in clinical contexts.

4 Comparative Analysis

4.1 Summary Table

4.2 Critical Analysis

Common strengths. The surveyed literature demonstrates convergent validity: across diverse methodologies—retrospective audits, template perturbation, representation probing, and large-scale vignette studies—a consistent finding emerges that AI systems produce systematically disparate outputs for different demographic groups. The most rigorous studies [15, 5] employ large-scale designs with multiple demographic dimensions and multi-model comparisons, yielding high statistical power. Methodological maturity is increasing: the field is moving from single-model case studies [2, 4] toward standardised, reproducible frameworks [5, 16].

Common limitations. Three limitations recur across virtually all reviewed papers:

1. **U.S.-centric scope.** All 14 papers evaluate U.S.-based demographic categories exclusively or primarily. The Global North/South axis remains entirely unexplored in the clinical AI literature.
2. **Limited model diversity.** Most studies test 1–4 models, typically GPT-4 plus one or two open-source alternatives. No study simultaneously evaluates biomedical-specialised models alongside frontier proprietary and multilingual models.
3. **Explicit vs. implicit demographic signals.** Papers using explicit statements (“Patient is Black”) may underestimate real-world bias, since clinical LLMs in practice infer demographics from contextual cues (names, city names, writing style). Only Gourabathina et al. [1] and Wan et al. [10] use implicit name-based signals, and neither extends to geographic identity.

Methodological trends. The field is evolving from small-scale case studies toward large-scale systematic evaluations [15]. LLM-as-annotator pipelines are increasingly adopted for scalable evaluation [7]. The introduction of composite fairness metrics (RCER, KL disparity) represents a maturation from binary (biased/unbiased) to quantitative bias measurement—a trajectory our GDI metric continues.

Scalability. Studies with explicit demographic injection [5, 14] scale readily to large-scale experiments but may underestimate bias that operates through implicit inference. Studies with name-based perturbation [10, 1] are more ecologically valid but require larger sample sizes to achieve statistical power, since the signal is noisier. Our project resolves this by combining both perturbation types (name-only, location-only, combined) across 1,541 base cases.

5 Research Gaps and Opportunities

The surveyed literature reveals five critical gaps that our project is positioned to address.

Gap 1: Absence of the Global South as an evaluation axis. Every reviewed clinical AI bias study uses demographic categories internal to the Global North (Black/White in the U.S., AAE dialect, insurance status). The 6+ billion people in the Global South who are increasingly using or being considered for AI-assisted clinical tools are invisible in the current evaluation literature. Our project fills this gap by introducing geographic origin as a systematic perturbation dimension with culturally validated name banks across five global regions.

Gap 2: Single-model evaluation. With limited exceptions [15, 4], reviewed papers test a single model or narrow pairs. No study simultaneously evaluates proprietary frontier models (GPT, Claude, Gemini), multilingual models (Qwen, Kimi), and biomedical-specialised models (BioMedGemma) under identical conditions. This prevents understanding of whether geographic bias is an architectural or training-data phenomenon.

Gap 3: Implicit name-based geographic inference. Existing studies either use explicit demographic statements or U.S.-centric name pairs. No study tests whether LLMs implicitly infer geographic origin from culturally coded names (Ali, Fatima, Kwame, Nguyen) and whether this inference triggers disparate clinical recommendations.

Gap 4: Absence of a geographic fairness metric. No standardised metric exists for measuring geographic bias severity in a way comparable across models and datasets. Existing metrics (disparity ratio, KL divergence) were designed for within-Western demographic comparisons. Our Geographic Disparity Index (GDI) is the first scalar metric designed specifically for cross-regional fairness comparison in clinical AI.

Gap 5: Intersectional geographic–gender analysis. While intersectionality has been theorised in AI fairness [9], no clinical AI paper evaluates the interaction between geographic origin and gender (e.g., whether Global South women face compounded disparities). Our project includes this as a secondary analysis dimension.

6 Relevance to Your Project

Adopted techniques.

- *Semantic-preserving perturbation:* adopted directly from Gourabathina et al. [1] and Wan et al. [10], extended to geographic identity (name + city pairs across 5 global regions and ~200 culturally validated names).
- *LLM-as-annotator pipeline:* adopted from Zheng et al. [7], using LLaMA-3-8B to extract binary MANAGE/VISIT/RESOURCE labels from ~75,000 model responses

at scale.

- *Three-question annotation schema*: adopted directly from Gourabathina et al. [1].
- *Statistical framework*: ANOVA with Bonferroni correction and paired t -tests with FDR correction, following Pfohl et al. [5] and Poulain et al. [14].

Baselines. Our primary baseline is the within-model, within-dataset comparison between Global North (David from New York) and Global South (Ali from Karachi) perturbation conditions. Secondary baselines draw on Gourabathina et al. [1] (Black/White name disparity in U.S. context) and Omar et al. [15] (sociodemographic disparity in ED settings).

Relevant datasets from the literature.

- *OncQA* [17]: used in Gourabathina et al. [1]; 61 filtered cancer patient-clinician message pairs in our study.
- *AskDocs* ($r/AskDocs$): used in Gourabathina et al. [1]; 147 informal medical queries.
- *USMLE-MedQA* [18]: reviewed in Thirunavukarasu et al. [13] and Hager et al. [6]; 1,333 filtered vignettes across 12 specialties.

Evaluation metrics. Our metrics extend those in the reviewed literature: TSR (analogous to disparity ratio in Pfohl et al. [5]), RCR and RCER (extending Gourabathina et al.’s annotation schema), and the novel GDI (addressing Gap 4 above). Significance thresholds follow the Bonferroni-corrected $p < 0.005$ standard used in Omar et al. [15].

Extension beyond existing work. Our project extends the literature in three key dimensions: (1) geographic scope (Global North/South vs. within-U.S. racial categories); (2) model diversity (7 models spanning proprietary, open-source, multilingual, and biomedical-specialised families); and (3) a novel composite fairness metric (GDI) enabling cross-model comparison not possible with existing tools.

7 Conclusion

This literature review has surveyed 14 papers at the intersection of clinical AI bias, LLM evaluation, and geographic/cultural fairness in NLP. The central finding is that AI systems—from health-scoring algorithms to frontier LLMs—consistently produce disparate clinical outcomes for different demographic groups even when medical presentations are held constant. Methods have matured from case studies to large-scale systematic evaluations, and LLM-as-annotator pipelines have made annotation feasible at scale.

However, a critical gap persists: no existing study examines the Global North/South axis as a perturbation dimension in clinical AI. All reviewed clinical bias work operates within U.S.-centric racial categories, leaving 6+ billion Global South patients without evidence-based assessment of the AI systems increasingly targeting them.

Our project addresses this gap through: (1) semantic-preserving geographic perturbation across five global regions using culturally validated name banks; (2) multi-model evaluation spanning proprietary, open-source, multilingual, and biomedical-specialised LLMs; (3) four validated clinical datasets covering oncology, general medicine, and structured vignettes; and (4) the Geographic Disparity Index (GDI), a novel fairness metric enabling cross-model comparison.

Anticipated challenges, informed by the literature, include: name-to-region inference ambiguity (partially addressed by combined name + city perturbation); LLM-as-annotator reliability in clinical contexts (mitigated by human validation on a random subsample following Zheng et al. [7]); and the risk that name-based inferences conflate geographic origin with other correlated variables (addressed through intersectional analysis of the geographic $\times$ gender interaction). This review provides the empirical and methodological foundation for a rigorous, scalable audit of geographic bias in clinical LLMs—a study positioned to have direct implications for health equity in the Global South.

References

- [1] S. Gourabathina, T. Naous, and W. Xu, “The medium is the message: How non-clinical information shapes clinical decisions in LLMs,” in *Proc. ACM FAccT*, 2025.
- [2] T. Zack, E. Steinberg, R. Nori, J. Listgarten, J. Naik, C. Valliani, V. Sherr, T. Pierson, B. Mach, and N. Shah, “Assessing the potential of GPT-4 to perpetuate racial and gender biases in health care: A model evaluation study,” *The Lancet Digital Health*, vol. 6, no. 1, pp. e12–e22, 2024.
- [3] V. Hofmann, J. Kalluri, D. Jurafsky, and S. King, “AI generates covertly racist decisions about people based on their dialect,” *Nature*, vol. 633, pp. 147–154, 2024.
- [4] J. A. Omiye, J. C. Lester, S. Spichak, V. Rotemberg, and R. Daneshjou, “Large language models propagate race-based medicine,” *npj Digital Medicine*, vol. 6, art. 195, 2023.
- [5] S. Pfohl *et al.*, “A toolbox for surfacing health equity harms and biases in large language models,” *Nature Medicine*, vol. 30, pp. 3333–3348, 2024.
- [6] P. Hager, F. Jungmann, R. Holland, K. Bhatt, E. Hubber, B. Knauer, F. Jakab, N. Bhatt, A. Lottermoser, and D. Rueckert, “Evaluation and mitigation of the limitations of large language models in clinical decision-making,” *Nature Medicine*, vol. 30, pp. 2613–2622, 2024.
- [7] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Li, Z. Li, D. Li, E. P. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica, “Judging LLM-as-a-judge with MT-Bench and chatbot arena,” in *Proc. NeurIPS*, vol. 36, 2023.
- [8] Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, “Dissecting racial bias in an algorithm used to manage the health of populations,” *Science*, vol. 366, no. 6464, pp. 447–453, 2019.
- [9] S. L. Blodgett, S. Barocas, H. Daumé III, and H. Wallach, “Language (technology) is power: A critical survey of ‘bias’ in NLP,” in *Proc. ACL*, pp. 5454–5476, 2020.
- [10] Y. Wan, G. Pu, J. Sun, A. Garimella, K.-W. Chang, and N. Peng, “‘Kelly is a warm person, Joseph is a role model’: Gender biases in LLM-generated reference letters,” in *Findings of EMNLP*, pp. 3730–3748, 2023.
- [11] R. Manvi, S. Khanna, M. Burke, D. Lobell, and S. Ermon, “Large language models are geographically biased,” *arXiv preprint arXiv:2402.02680*, 2024.
- [12] F. Faisal and A. Anastasopoulos, “Geographic and geopolitical biases of language models,” in *Proc. Workshop on Multilingual Representation Learning (MRL) at EMNLP*, pp. 115–129, 2023.
- [13] A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, “Large language models in medicine,” *Nature Medicine*, vol. 29, pp. 1930–1940, 2023.
- [14] R. Poulain, H. Fayyaz, and R. Beheshti, “Bias patterns in the application of LLMs for clinical decision support: A comprehensive study,” *arXiv preprint arXiv:2404.15149*, 2024.
- [15] M. Omar, S. Soffer, R. Agbareia, N. L. Bragazzi, D. U. Apakama, C. R. Horowitz,

- A. W. Charney, R. Freeman, B. Kummer, B. S. Glicksberg, G. N. Nadkarni, and E. Klang, “Sociodemographic biases in medical decision making by large language models,” *Nature Medicine*, vol. 31, pp. 1873–1881, 2025.
- [16] A. Parrish, A. Chen, N. Nangia, V. Padmakumar, J. Phang, J. Thompson, P. M. Htut, and S. Bowman, “BBQ: A hand-built bias benchmark for question answering,” in *Findings of ACL*, pp. 2086–2105, 2022.
- [17] E. Chen, D. Aggarwal, J. Lee, K. Ash, B. Chen, and N. Shah, “The impact of using an AI chatbot to respond to patient messages,” *arXiv preprint arXiv:2310.17703*, 2023.
- [18] D. Jin, E. Pan, N. Oufattole, W.-H. Weng, H. Fang, and P. Szolovits, “What disease does this patient have? A large-scale open domain question answering dataset from medical exams,” *arXiv preprint arXiv:2009.13081*, 2020.
- [19] S. M. Petterson, W. R. Liaw, R. L. Phillips, D. L. Rabin, D. S. Meyers, and A. W. Bazemore, “Projecting US primary care physician workforce needs: 2010–2025,” *Annals of Family Medicine*, vol. 10, no. 6, pp. 503–509, 2012.

Table 1: Comparison of Reviewed Approaches

Paper	Technique	Dataset(s)	Key Finding	Limitations
Obermeyer et al. [8]	Retrospective cohort audit	43,539 EHR patients	50% less Black enrolment in care mgmt	Single algorithm; single institution
Zack et al. [2]	Clinical vignette testing	4 clinical vignettes	GPT-4 reproduces race-based medical myths	4 cases; single model
Omiye et al. [4]	Multi-model topic audit	10 topics $\times$ 4 LLMs	All models perpetuate racial medical claims	Binary evaluation only
Omar et al. [15]	Large-scale demographic perturbation	1,000 ED cases $\times$ 32 variants	Disparities for Black/unhoused/LGBTQ+ patients	U.S.-centric; no geographic axis
Gourabathina et al. [1]	Semantic-preserving name perturbation	OncQA, AskDocs	Black-name patients get more self-management	U.S. names only; single model
Hofmann et al. [3]	Dialect substitution	AAE/SAE corpus	Covert racism via dialect proxy	U.S. dialect only
Manvi et al. [11]	Geographic knowledge probing	World Bank/UN, 170+ countries	GDP-correlated geographic bias in LLMs	Not clinical domain
Faisal & Anastasopoulos [12]	Representation probing	Wikipedia, 193 countries	Geopolitical bias in LM representations	Not downstream tasks
Thirunavukarasu et al. [13]	Systematic literature review	110+ papers; USMLE-MedQA	LLMs degrade on non-Western inputs	No geographic bias audit
Hager et al. [6]	Structured failure evaluation	300 expert clinical tasks	Chain-of-thought improves accuracy, not bias	U.S./EU-centric tasks
Pfohl et al. [5]	Equity toolbox (457 scenarios)	457 scenarios $\times$ 8 variants	Insurance/race drive largest disparities	No geographic axis
Poulain et al. [14]	KL-divergence perturbation	MedQA, ClinicalNLP	Fine-tuning amplifies bias by 23%	No geographic signals
Wan et al. [10]	Name-based perturbation	100 prompts $\times$ 5 LLMs	Female names receive communal language	Binary gender; U.S.-centric
Parrish et al. [16]	Counterfactual QA benchmark	58,492 BBQ examples	Models rely on stereotypes under ambiguity	Classification only; no clinical